

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Vinorelbine Injection

VINELBINE

SAFETY WARNINGS: Myelosuppression

Severe myelosuppression resulting in serious infection, septic shock, hospitalization and death may occur.

Decrease the dose or withhold vinorelbine in accord with recommended dose modifications.

DESCRIPTION:

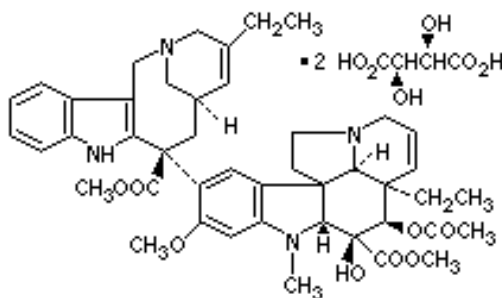
Vinorelbine is a semisynthetic vinca alkaloid, derived from vinblastine, with a broad spectrum of antitumour activity. It is a derivative of vinblastine and has structural modifications distinctly different from other vinca alkaloids and a spectrum of activity similar to either vincristine or vinblastine. However, vinorelbine may have less neurotoxicity than vincristine.

Vinorelbine Injection is clear colourless to pale yellow solution.

COMPOSITION:

Each ml contains:

Vinorelbine tartrate equivalent to Vinorelbine	10 mg
Water for Injection USP	q.s.

CHEMICAL STRUCTURE:

Vinorelbine tartrate is a white to yellow or light brown amorphous powder. Chemically it is 3', 4'-didehydro-4'-deoxy-*C'*-norvincal leukoblastine [R-(*R**,*R**)-2,3-dihydroxybutanedioate (1:2)(salt)]. The aqueous solubility is >1,000 mg/ml in distilled water. It has a molecular formula of $C_{45}H_{54}N_4O_8 \cdot 2C_4H_6O_6$ and molecular weight of 1,079.12

PHARMACOLOGY:**Pharmacodynamic:**

Pharmacotherapeutic class: Cytotoxic antineoplastic belonging to the vinca alkaloid family.
ATC Code: L01CA04 (L - Antineoplastics and immunomodulators)

Vinorelbine is an antineoplastic drug of the vinca alkaloid family but unlike all the other vinca alkaloids, the cathartine moiety of vinorelbine has been structurally modified. At the molecular level, it acts on the dynamic equilibrium of tubulin in the microtubular apparatus of the cell. Vinorelbine inhibits tubulin polymerization and binds preferentially to mitotic microtubules, affecting axonal microtubules at high concentrations only. The induction of tubulin spiralization is less than that produced by vincristine.

Vinorelbine blocks mitosis at G2-M, causing cell death in interphase or at the following mitosis.

Pharmacokinetics:

Pharmacokinetic parameters of vinorelbine were evaluated in blood.

Distribution

The steady-state volume of distribution is large, on average 21.2 l/kg (range: 7.5-39.7 l/kg), which indicates extensive tissue distribution. Binding to plasma protein is low (13.5%). However, vinorelbine binds strongly to blood cells and especially to platelets (78%).

There is significant uptake of vinorelbine in the lungs, as assessed by surgical lung biopsies which showed concentrations up to 300-fold higher than in serum. Vinorelbine is not found in the central nervous system.

Biotransformation

All metabolites of vinorelbine are formed by CYP 3A4 isoform of cytochromes P450, except 4-O-deacetylvinorelbine likely to be formed by carboxylesterases. 4-O-deacetylvinorelbine is the only active metabolite and the main one observed in blood.

No sulphonate or glucuronic conjugates are found.

Elimination

The mean terminal half-life of vinorelbine is around 40 hours. Blood clearance is high, approaching hepatic blood flow, and is 0.72 l/h/kg on average (range: 0.32 – 1.26 l/h/kg).

Renal elimination is low (< 20% of the intravenous dose administered) and consists mostly in parent compound. Biliary excretion is the predominant elimination route of both metabolites and unchanged vinorelbine, which is the main recovered compound.

Special patient groups

Renal and liver impairment

The effects of renal dysfunction on vinorelbine disposition have not been assessed. However, dose reduction in case of reduced renal function is not indicated due to the low renal elimination.

Mean total clearance in these two subsets of patients was similar to that in patients with normal hepatic function. Therefore, the pharmacokinetics of vinorelbine is not modified in patients presenting moderate or severe liver impairment. Nevertheless, as a precautionary measure a reduced dose of 20 mg/m² and close monitoring of haematological parameters is recommended in patient with severe liver impairment.

Elderly Patients

A study with vinorelbine in elderly patients (≥ 70 years) with NSCLC demonstrated that pharmacokinetics of vinorelbine were not influenced by age. However, since elderly patients are frail, caution should be exercised when increasing the dose of vinorelbine.

Pharmacokinetic/pharmacodynamic relationships

A strong relationship has been demonstrated between vinorelbine blood exposure and of leucocytes or PMNs decreases.

ROUTE OF ADMINISTRATION:

Intravenous injection by slow bolus (6-10 minutes) after appropriate dilution.

INDICATIONS:

- Non-small cell lung cancer
- Advanced breast cancer

CONTRAINDICATIONS:

This medicine is contra-indicated in the following cases:

- Known hypersensitivity to vinorelbine or other vinca alkaloids, or to any of the constituents.
- Neutrophil count $< 1500/\text{mm}^3$ or severe infection current or recent (within 2 weeks).
- Platelet count $< 100000/\text{mm}^3$.
- Lactation.
- In combination with yellow fever vaccine.

ADVERSE EFFECTS:

Adverse reactions reported as more than isolated cases are listed below, by system organ class and by frequency. Frequencies are defined as: very common ($> 1/10$), common ($> 1/100$, $< 1/10$), uncommon ($> 1/1,000$, $< 1/100$), rare ($> 1/10,000$, $< 1/1,000$), very rare ($< 1/10,000$), according to the MedDRA frequency convention and system organ classification.

The most commonly reported adverse drug reactions are bone marrow depression with neutropenia, anaemia, neurologic disorders, gastrointestinal toxicity with nausea, vomiting, stomatitis and constipation, transient elevations of liver function tests, alopecia and local phlebitis.

Additional Adverse reactions from Post Marketing experience have been added according to the MedDRA classification with the frequency Not known.

Detailed information:

Reactions were described using the W.H.O classification (grade 1=G1; grade 2=G2; grade 3=G3; grade 4=G4; grade 1-4=G1-4; grade 1-2=G1-2; grade 3-4=G3-4).

Infections and infestations

Common: bacterial, viral or fungal infection at different localization (respiratory, urinary, GI tract) mild to moderate and usually reversible with an appropriate treatment.

Uncommon: Severe sepsis with other visceral failure. Septicaemia

Very rare: Complicated septicaemia and sometimes fatal.

Not known: Neutropenic sepsis.

Blood and lymphatic system disorders

Very Common: Bone marrow depression resulting mainly in neutropenia (G3: 24.3%; G4: 27.8%), reversible within 5 to 7 days and noncumulative over time. Anaemia (G3-4: 7.4%)

Common: Thrombocytopenia (G3-4: 2.5%) may occur but are seldom severe.

Not known: Febrile neutropenia, Pancytopenia.

Immune system disorders

Not known: Systemic allergic reactions as anaphylaxis, anaphylactic shock or anaphylactoid type reaction.

Endocrine disorders

Not known: Inappropriate antidiuretic hormone secretion (SIADH).

Metabolism and nutrition disorders

Rare: Severe hyponatraemia.

Not known: Anorexia.

Nervous system disorders

Very Common: Neurologic disorders (G 3-4: 2.7%) including loss of deep tendon reflexes.

Weakness of the lower extremities has been reported after a prolonged chemotherapy.

Uncommon: Severe paresthesias with sensory and motor symptoms. These effects are generally reversible.

Cardiac disorders

Rare: Ischemic heart disease (angina pectoris, myocardial infarction, occasionally fatal).

Very rare: Tachycardia, palpitations and heart rhythm disorders.

Vascular disorders

Uncommon: Hypotension, hypertension, flushing and peripheral coldness.

Rare: Severe hypotension, collapse.

Respiratory system, thoracic and mediastinal disorders

Uncommon: Dyspnoea and bronchospasm may occur in association with vinorelbine treatment as with other vinca alkaloids.

Rare: Occasionally fatal interstitial lung disease.

Gastrointestinal disorders

Very Common: Stomatitis (G1-4: 15% with vinorelbine as single agent). Nausea and vomiting (G1-2: 30.4% and G3-4: 2.2%). Anti-emetic therapy may reduce their occurrence.

Constipation is the main symptom (G3-4: 2.7%) which rarely progresses to paralytic ileus with vinorelbine as single agent and (G3-4: 4.1%) with the combination of vinorelbine and other chemotherapeutic agents.

Common: Diarrhoea usually mild to moderate.

Rare: Paralytic ileus, treatment may be resumed after recovery of normal bowel mobility. Pancreatitis.

Hepatobiliary disorders

Very common: Transient elevations of liver function tests (G1-2) without clinical symptoms were reported (SGOT in 27.6% and SGPT in 29.3%).

Skin and subcutaneous tissue disorders

Very Common: Alopecia, usually mild in nature, may occur (G3-4: 4.1% with vinorelbine as single chemotherapeutic agent).

Rare: Generalized cutaneous reactions.
Not known: Palmo-plantar erythrodysesthesia.

Musculoskeletal and connective tissue disorders

Common: Arthralgia including jaw pain and myalgia.

General disorders and administration site conditions

Very common: Reactions at the injection site may include erythema, burning pain, vein discoloration and local phlebitis (G3-4: 3.7% with vinorelbine as single chemotherapeutic agent).

Common: Asthenia, fatigue, fever, pain at different locations including chest pain and pain at the tumour site have been experienced by patients receiving vinorelbine therapy.

Rare: Local necrosis. Proper positioning of the intravenous needle or catheter and bolus injection followed by liberal flushing of the vein can limit these effects.

Reporting of side effects

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

INTERACTIONS WITH OTHER MEDICAMENTS:

Interactions common to all cytotoxics:

Concomitant use contraindicated

+ Yellow fever vaccine: Risk of fatal generalised vaccine disease. Concomitant use not recommended

+ Live attenuated vaccines (for yellow fever vaccine, see Concomitant use contraindicated): Risk of generalised vaccine disease, possibly fatal. This risk is increased in patients already immunodepressed by their underlying disease. It is recommended to use an inactivated when exists (poliomyelitis).

+ Phenytoin (and by extrapolation, fosphenytoin): Risk of exacerbation of convulsions resulting from the decrease of phenytoin digestive absorption by cytotoxic drug or loss of efficacy of the cytotoxic drug due to increased hepatic metabolism by phenytoin or fosphenytoin.

Interaction with special precaution for use:

+ Oral anticoagulant: There is an increased thrombotic and haemorrhagic risk in case of tumoral diseases. There is an eventuality of interaction between oral anticoagulants and anticancer chemotherapy. Increased frequency of the INR (International Normalised Ratio) monitoring is required.

+ Macrolides (clarithromycin, erythromycin, telithromycin): Risk of increased toxicity of the anti-mitotic agent due to a reduction in its hepatic metabolism by clarithromycin, erythromycin or telithromycin. Close clinical and laboratory monitoring. Possibly, use an alternative antibiotic.

+ Cobicistat: Increased neurotoxicity of the antimitotic due to a reduction in its hepatic metabolism by cobicistat. Close clinical monitoring and possible adjustment of dosage of the anti-mitotic agent.

Concomitant use to take into consideration:

+ Immunosuppressive medicines (ciclosporine, tacrolimus, everolimus, sirolimus): Excessive immunodepression with risk of lymphoproliferation.

Interactions specific to vinca-alkaloids:

Concomitant use not recommended

+ Itraconazole, posaconazole, ketoconazole: Increased neurotoxicity of vinca-alkaloids due to the decrease of their hepatic metabolism.

Interaction with special precaution for use:

+ Protease inhibitors: Increased toxicity of the antimitotic agent due to the decrease of its hepatic metabolism by protease inhibitors. Close clinical monitoring and eventually adaptation of the antimitotic agent dosage is required.

Concomitant use to take into consideration:

+ Mitomycin C: risk of increased pulmonary toxicity of mitomycin and vinca-alkaloids.

As vinca-alkaloids are known as substrates for P-glycoprotein, and in the absence of specific study, caution should be exercised when combining vinorelbine with strong modulators of this membrane transporter.

Interactions specific to vinorelbine:

As CYP3A4 is mainly involved in the metabolism of vinorelbine, combination with strong inhibitors of this isoenzyme could increase blood concentration of vinorelbine and combination with strong inducers of this isoenzyme could decrease blood concentration of vinorelbine.

The combination of vinorelbine with other drugs with known bone marrow toxicity is likely to exacerbate the myelosuppressive adverse effects. There is no mutual pharmacokinetic interaction when combining vinorelbine with cisplatin over several cycles of treatment. However, the incidence of granulocytopenia associated with vinorelbine use in combination with cisplatin is higher than associated with vinorelbine single agent.

WARNINGS AND PRECAUTIONS:

WARNINGS

Vinelbine Injection should be administered under the supervision of a physician experienced in the use of chemotherapy.

Since inhibition of the hematopoietic system is the main risk associated with Vinelbine Injection, close haematological monitoring should be undertaken during treatment (determination of hemoglobin level and the leukocyte, neutrophil and platelet counts on the day of each new administration).

The dose limiting adverse reaction is mainly neutropenia. This effect is non-cumulative, having its nadir between 7 and 14 days after the administration and is rapidly reversible within 5 to 7 days. If the neutrophil count is below $1500/\text{mm}^3$ and/or the platelet count is below $100000/\text{mm}^3$, then the treatment should be delayed until recovery.

If patients present signs or symptoms suggestive of infection, a prompt investigation should be carried out.

PRECAUTIONS

Special care should be taken when prescribing for patients with history of ischaemic heart disease.

The pharmacokinetics of Vinelbine Injection is not modified in patients presenting moderate or severe liver impairment.

For dosage adjustment in this specific patient group, refer to section “DOSAGE AND ADMINISTRATION”.

As there is low level of renal excretion there is no pharmacokinetic rationale for reducing Vinelbine Injection dose in patients with impaired kidney function. Refer to section “DOSAGE AND ADMINISTRATION”.

Use of this medicinal product in combination with a live attenuated vaccine is not recommended (see CONTRAINDICATIONS for the yellow fever vaccine).

Caution is recommended when Vinelbine Injection is used at the same time as potent cytochrome CYP3A4 inhibitors or inducers. Hence, taking this medicinal product with phenytoin, fosphenytoin, itraconazole, ketoconazole or posaconazole is not recommended (see section INTERACTIONS).

All contact with the eyes should be strictly avoided: there is a risk of severe irritation and even corneal ulceration if the drug is sprayed under pressure. Immediate washing of the eye with sodium chloride 9 mg/ml (0.9%) solution for injection should be undertaken if any contact occurs.

Vinelbine Injection should not be administered concomitantly with radiotherapy where the field includes the liver.

Increased caution is required in Japanese patients as cases of interstitial lung disease have been reported more frequently in this population.

Mutagenic, and carcinogenic potential

Vinorelbine induced chromosome damages but was not mutagenic in Ames test. It is assumed that vinorelbine can cause mutagenic effects (induction aneuploidy and polyploidy) in man.

Toxicity to reproduction

In animal reproductive studies, vinorelbine was embryo-foeto-lethal and teratogenic.

Safety Pharmacology

No haemodynamic effects were found in dogs receiving vinorelbine at maximal tolerated dose; only some minor, non-significant disturbances of repolarisation were observed as with other vinca alkaloids tested. No effect on the cardiovascular system was observed in primates receiving repeated doses of vinorelbine over 39 weeks.

Pregnancy:

There are inadequate data on the use of vinorelbine in pregnant women. In reproductive studies conducted in animals, vinorelbine was embryotoxic and teratogenic.

Based on the results of these animal studies and the pharmacological action of the medicinal product there is a potential risk of embryonic and foetal abnormalities. Vinorelbine must not

be used during pregnancy unless the expected individual benefit manifestly exceeds the potential risks. If a patient becomes pregnant during treatment she must be informed of the risks to the unborn child and monitored carefully. The possibility of genetic counselling should also be considered.

Women of child-bearing potential:

Women of child-bearing potential must use effective contraception during treatment and up to 3 months after treatment.

Lactation:

It is unknown whether vinorelbine is excreted in human breast milk. The excretion of vinorelbine in milk has not been studied in animal studies. A risk to the suckling cannot be excluded therefore breast feeding must be discontinued before starting treatment with vinorelbine.

Fertility:

Men being treated with vinorelbine are advised not to father a child during and up to 3 months after treatment. Prior to treatment, advice should be sought for conserving sperm due to the chance of irreversible infertility as a consequence of treatment with vinorelbine.

Effects on Ability to Drive and use Machines

No studies on the effects on the ability to drive and use machines have been performed but on the basis of the pharmacodynamic profile vinorelbine does not affect the ability to drive and use machines. However, caution is necessary in patient treated with vinorelbine considering some adverse effects of the drug.

DOSAGE AND ADMINISTRATION:

Strictly intravenous administration after appropriate dilution. Intra-theal administration of vinorelbine may be fatal. See section INSTRUCTIONS FOR USE AND HANDLING.

As a single agent, the usual dose is 25 to 30 mg/m² administered weekly.

In combination chemotherapy, the usual dose (25-30 mg/m²) is usually maintained, while the frequency of administration is reduced e.g. day 1 and 5 every 3 weeks or day 1 and 8 every 3 weeks according to treatment protocol.

Vinorelbine may be administered by slow bolus (6-10 minutes) after dilution in 20-50 ml of Sodium chloride 9 mg/ml (0.9%) solution for injection or in 5% glucose solution for injection. Administration should always be followed with at least 250 ml of an isotonic solution infusion to flush the vein.

Administration in the elderly: Clinical experience has not identified relevant differences in the elderly with regards to response rate, although greater sensitivity of some these patients cannot be excluded.

Age does not modify the pharmacokinetics of vinorelbine.

Administration in patients with liver insufficiency: The pharmacokinetics of vinorelbine is not modified in patients presenting moderate or severe liver impairment. Nevertheless, as a

precautionary measure a reduced dose of 20 mg/m² and close monitoring of haematological parameters is recommended in patient with severe liver impairment.

Administration in patients with renal insufficiency: Given the minor renal excretion, there is no pharmacokinetic justification for reducing the dose of vinorelbine in patients with renal insufficiency.

Administration in children: Safety and efficacy in children have not been established and administration is therefore not recommended.

OVERDOSE AND TREATMENT:

Symptoms

Overdosage with vinorelbine could produce bone marrow hypoplasia sometimes associated with infection, fever and paralytic ileus.

Emergency procedure

General supportive measures together with blood transfusion, growth factors and broad-spectrum antibiotic therapy should be instituted as deemed necessary by the physician.

Antidote

There is no known antidote for overdosage of vinorelbine.

STORAGE CONDITION:

Store the vials under refrigeration at 2° to 8° C (36° to 46° F) in the carton. Protect from light. Do not freeze. Diluted vinorelbine (in a 5% dextrose solution) may be used for up to 24 hours under normal room light when stored in polypropylene syringes or polyvinyl chloride bags at 5° to 30° C (41° to 86° F).

INCOMPATIBILITIES:

Vinorelbine should not be diluted in alkaline solutions (risk of precipitation). This medicinal product must not be mixed with other medicinal product except those mentioned in "INSTRUCTIONS FOR USE AND HANDLING".

SHELF LIFE:

24 months

HANDLING AND DISPOSAL:

INSTRUCTIONS FOR USE AND HANDLING

The preparation and administration of vinorelbine should be carried out by trained staff. Suitable eye protection, disposable gloves, face mask and disposable apron should be worn. Eventual spillage or leakage should be mopped up.

In case of contact with the eye, immediate liberal washing of the eye with sodium chloride 0.9% solution for injection should be undertaken.

In case of accidental skin projection, proceed with a cleaning with water and mild soap followed by a thorough washing with water.

On completion, any exposed surface should be thoroughly cleaned, and hands and face washed. There is no content / container incompatibility between vinorelbine and neutral glass bottle, PVC bag, vinyl acetate bag or infusion set with PVC tubing.

It is recommended to infuse vinorelbine over 6-10 minutes after dilution in 20-50 ml of sodium chloride 0.9% solution for injection or in glucose solution for injection 5%. After administration the vein should be thoroughly flushed with at least 250 ml of isotonic solution.

Vinorelbine must be given strictly intravenously: it is very important to make sure that the cannula is accurately placed in the vein before starting to infuse vinorelbine. If the drug extravasates into the surrounding tissue during the administration considerable local irritation may occur. In this case, the administration should be stopped, the vein flushed with normal saline, and the extravasated product should be removed, and the remaining dose administered in another vein. Application of mild heat facilitates product diffusion and seems to reduce risk of cellulitis. In case of extravasations, to reduce the risk of phlebitis IV glucocorticoids could be administered immediately. Pregnant women should be warned and avoid handling cytotoxic agents.

Before any administration, injection solution should be visually inspected so as to detect presence of particles or discoloration.

Any unused product or waste material should be disposed of in accordance with local requirements.

PRESENTATION:

VINELBINE 10 mg

1 ml in 5 ml type I Clear glass vial with 20 mm fluorotec stoppered rubber plug and 20 mm flip off seal.

VINELBINE 50 mg

5 ml type I Clear glass vial with 20 mm fluorotec stoppered rubber plug and 20 mm flip off seal.

REFERENCES:

Prescribing Information, Navelbine (Vinorelbine) Solution for Injection in vial, Pierre Fabre Medicament, France, November 2016.

MANUFACTURER INFORMATION:

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